

My research sits at the intersection of mathematics, data science, and computational social science, where I use topological data analysis (TDA) and network science to uncover the structures of complex social systems. Both approaches reveal structure that is not visible locally, whether it is a gap in a data set or a pattern of collective behavior. My work spans both the application of these tools and the development of their mathematical foundations.

I have used TDA to study geospatial resource access [9, 21] and to identify topological holes in embeddings of scientific concepts [12]. I have also formulated and analyzed agent-based opinion models to study collective behavior in networks, extending classical frameworks by incorporating richer social-interaction mechanisms [10, 11, 20]. My work in these areas also has important theoretical components. In TDA, I study persistence modules [13, 14] (the algebraic backbone of persistent homology) and address fundamental questions about their structure. In network science, I analytically study the dynamics of the models I formulate, proving convergence guarantees and characterizing their long-term behavior.

I am deeply committed to mentoring students, which is a central component of my research program. I have mentored both undergraduate and master’s students on research projects; the first two authors of [20] were undergraduate mentees (co-mentored with Mason A. Porter).

1 Topological Data Analysis and Applications

TDA [1] uses tools derived from algebraic topology to quantify the shape of data concisely and robustly. One of the primary tools in TDA is *persistent homology* (PH) [17], which is concerned with the “persistence” of holes in data. Given a data set (often a point cloud), one constructs a *filtration*, which is a nested sequence $\mathcal{K}_1 \subseteq \mathcal{K}_2 \subseteq \dots \subseteq \mathcal{K}_r$ of simplicial complexes¹ that approximates the data set across different scales. A common choice is the Vietoris–Rips (VR) filtration, which at scale r includes a simplex whenever all of its vertices are within distance r of each other. As the scale grows, new holes form (are born) and existing holes fill in (die). This information is summarized in a *persistence diagram*, which records the scales at which each hole is created and fills in. For each hole, the birth simplex is the simplex whose addition creates it and the death simplex is the simplex whose addition fills it in.

I have used TDA to analyze geospatial data and assess the accessibility of resources over geographical regions (see Section 1.1). More recently, I have applied TDA to the science of science, studying topological representations of scientific knowledge (see Sections 1.2 and 1.3). I plan to continue in this area, extending my work to new questions about how scientific structures evolve (see Section 4.2).

1.1 Persistent Homology for Resource Coverage

Access to public resources is unevenly distributed, and quantifying that unevenness motivates a notion of “underserved” that does not depend on an arbitrary threshold distance. PH offers a natural framework to do this. A hole in a coverage filtration identifies a region that remains unserved up to a given scale, and the death of that hole measures how severely it is underserved.

In [9], my collaborators and I studied the accessibility of polling sites in six cities: Atlanta, Chicago, Jacksonville, Los Angeles, New York City, and Salt Lake City. For each city, we constructed a weighted VR filtration that incorporates travel times to and waiting times at polling sites. We used the resulting persistence diagrams to identify holes in coverage and quantify their severity. The associated death simplices allow us to map those holes back to specific geographic locations, making the analysis both quantitative and visual. In related work [21], my collaborators and I used a PH-based approach that incorporates resource quality (e.g., ratings of public parks) for assessing accessibility.

1.2 Persistent Homology for Science of Science

Science advances by filling gaps, but systematically identifying where those gaps lie can be difficult. In [12], James A. Evans and I constructed embedding spaces of scientific concepts that encode semantic distances and used PH to identify the holes in these spaces. We computed the PH of each concept

¹A *simplicial complex* is a space built from simplices, which are vertices, edges, triangles, and higher-dimensional analogues.

embedding using a VR filtration and, as in Section 1.1, used the death simplex of each hole to identify the specific concepts that form it. We counted a hole as filled if a work combines all of those concepts. We found that works filling such holes tend to attract disproportionate attention, as measured by citation count and established measures of novelty.

In ongoing work, my collaborators² and I have built a VR filtration that encodes chronology. Vertices are concepts, added at their first appearance; edges are co-occurrences, added at the year of first joint use; and a higher-dimensional simplex appears once all of its edges do. PH then reveals topological holes in this conceptual space, where concepts are connected through indirect paths but have not yet been directly combined. We are analyzing how the process of creating holes relates to how they are filled in.

1.3 Mapping Scientific Knowledge

I am also working with other TDA tools beyond PH. I am currently using Mapper [19] — which visually summarizes data as a graph, whose nodes are clusters of nearby points and whose edges record overlap between those clusters — to study embeddings of scientific papers. In using Mapper, I seek to understand how disciplines are positioned relative to one another and to measure the distribution of highly cited papers across scientific space. I also seek to identify the boundary regions where fields transition and to identify “bridge” papers that sit in these regions and embody interdisciplinary research.

2 Network Science

Networks provide a natural framework for describing systems of locally interacting agents and studying how collective behavior emerges, from the sharing of opinions to the spread of pathogens in a population. My work focuses on how these interaction rules, together with network structure, shape the dynamics that unfold. I have worked primarily within opinion dynamics by formulating and analyzing mathematical models that describe how beliefs evolve from social interactions. In particular, I have worked with bounded-confidence models (BCMs) [15], a family of opinion models in which agents are receptive to neighbors whose opinions are sufficiently close, as determined by a “confidence bound” threshold parameter. I have generalized classical BCMs [6, 8] to incorporate richer social mechanisms. I am also studying how opinion dynamics interact with other social and biological processes.

My work in collective behavior on networks combines theoretical analysis with numerical simulations. My collaborators and I have proven convergence guarantees and characterized long-term behavior, which provide insight into the dynamics of our models and ground our numerical simulations. For instance, we studied “effective graphs”, which are time-dependent subgraphs of mutually receptive agents. In our generalized BCMs, we have proven that the effective graph stabilizes in finite time and that its surviving edges connect only agents who converge to the same opinion. The components of the effective graph help us determine the opinion clusters, which are groups of agents that converge to the same opinion.

Looking ahead, I plan to build on my work in studying collective behavior on networks by developing mean-field counterparts to agent-based models (see Section 4.3) and using an agent-based framework to study the dynamics of large-language-model agents (see Section 4.4).

2.1 Opinion Dynamics with Adaptive Mechanisms

I have studied several ways in which positive and negative interactions shape subsequent interactions in BCMs.

In [11], my collaborators and I formulated BCMs in which positive and negative interactions alter the agents’ subsequent receptiveness to each other. In our models, confidence bounds increase after positive interactions and decrease after negative ones. We found that adaptive confidence bounds reduce fragmentation (i.e., there are fewer final opinion clusters) compared to nonadaptive baselines and that agents can converge to the same opinion even when they are not mutually receptive in the limit — a phenomenon that does not occur in the baseline BCMs.

²James A. Evans (University of Chicago), Russell Funk (University of Minnesota), and Lori Ziegelmeier (Macalester College).

In [20], my collaborators³ and I formulated a BCM in which positive and negative interactions alter the probability of future interactions. In particular, we incorporated adaptive interaction probabilities between agents that increase after positive interactions and decrease after negative ones. This modeling choice encodes the tendency of people to communicate with those with whom they have previously interacted positively. We numerically observed that the effect of incorporating adaptive interaction probabilities depends on network density. For small confidence bounds, adaptive interaction probabilities shorten convergence times on dense networks and lengthen them on sparse ones.

I am working with students that I supervise (at the University of Chicago) on several ongoing projects that explore additional directions of how positive and negative interactions shape collective outcomes. With William Flowers (an undergraduate student), I formulated BCMs in which positive and negative interactions influence the magnitude of future compromise between agents. In our BCMs, positive interactions increase the extent to which agents adjust their opinions toward each other in subsequent encounters, while negative interactions decrease it. With Ziqian Zhao (a master's student), I formulated a BCM in which negative interactions can lead to network rewiring. If agents with sufficiently different opinions interact, they have some probability (associated with their opinion difference) of severing their connection and forming new connections with other agents.

2.2 Dynamics of Opinions with Interdependent Topics

People hold opinions on many topics at once, and those opinions are seldom independent. In [10], my collaborators and I formulated BCMs with multidimensional opinions in which receptivity on one topic depends on opinion differences across all topics. Numerically, we found that when initial opinions are correlated across topics, our models differ markedly — in both transient and steady-state behavior — from baselines in which each topic evolves independently.

2.3 Coupled Opinion and Disease Dynamics

Opinions and disease spread influence each other. Beliefs about a disease shape the behaviors that spread it, and the course of an outbreak reshapes those beliefs. In ongoing work, my collaborators⁴ and I formulated a coupled network model that combines a BCM with a compartmental disease model. Each agent has both an opinion and a disease state, which co-evolve. An agent's opinion influences its probability of infecting others or becoming infected, while changes in disease state lead to changes in opinion. We found that increasing agents' receptiveness leads to wider and faster disease spread across a variety of networks and that the presence of disease produces asymmetric final opinion distributions. On networks with two communities, with one risk-averse community and one risk-accepting community, we identified a confidence-bound threshold for disease outbreak in the risk-averse community.

3 Theoretical Foundations of Persistent Homology

Persistent homology rests on the theory of *persistence modules* [18], an algebraic structure that captures the information summarized in persistence diagrams. Given a filtration $\mathcal{K} = \{\mathcal{K}_i\}_{i=1}^r$, computing k th homology with coefficients in a field $\mathbb{F}$ yields a persistence module $H_k(\mathcal{K}; \mathbb{F})$, which consists of a sequence $\{H_k(\mathcal{K}_i; \mathbb{F})\}_{i=1}^r$ of vector spaces over $\mathbb{F}$, together with *structure maps* between them induced by inclusion. Field coefficients guarantee that persistence modules admit an *interval decomposition*, a result that leans upon Gabriel's classification of representations of type- A_n quivers [7].

Over a general ring, such as the integers, a persistence module of a filtration may not have an interval decomposition. In this setting, the persistence diagram of that filtration can depend on the choice of coefficient field [16]. My work establishes necessary and sufficient conditions for the existence of interval decompositions with coefficients in a principal ideal domain (PID), a class that includes the integers. A corollary of this result is that, in torsion-free settings, the persistence module of a filtration with integer coefficients has an interval decomposition if and only if its persistence diagram is independent of the

³Leila Thompsky and Yuexuan Wu were undergraduate students whom I co-supervised with Mason A. Porter.

⁴Yang Yang (The Ohio State University), Mason A. Porter (Nordita), and Joseph Tien (The Ohio State University).

choice of coefficient field.

In [14], Gregory Henselman-Petrusek and I studied finitely indexed persistence modules with coefficients in a PID. We proved the following theorem for these persistence modules.

Theorem 1. *A persistence module that is pointwise free and finitely generated admits an interval decomposition if and only if the cokernel of every structure map is free.*

We also developed a polynomial-time algorithm that produces an interval decomposition for such a persistence module (whenever one exists) or certifies its non-existence. In subsequent work [13], we generalized our result to persistence modules indexed by an arbitrary totally ordered poset category.

Looking ahead, we plan to generalize our work (see Section 4.1) to zigzag persistence modules, persistence modules that are not pointwise free, and persistence modules indexed by more general posets.

4 Research Plan

I will build on my existing research by extending my work in TDA and network science to new domains and by developing complementary mathematical frameworks.

4.1 Interval Decomposition of Persistence Modules over PIDs

In collaboration with Gregory Henselman-Petrusek, I seek to extend our program (Section 3) in several directions. First, we will develop a framework for zigzag persistence modules over PIDs and determine when they admit interval decompositions. Zigzag persistence [2] arises in settings where a filtration is not monotone, such as when considering time-dependent data. Second, we aim to move beyond free modules to general modules (allowing for torsion) over a PID. I expect torsion to be the principal obstruction, requiring case-by-case treatment according to the orders of the torsion elements involved, as well as a generalized definition of interval decomposition. Third, we will extend our interval-decomposition results to persistence modules indexed by more general posets. Although interval decompositions do not exist in the general multiparameter setting (even with field coefficients) [3], we intend to investigate indexing posets with additional structure, such as trees and other posets of bounded representation type, where a more complete characterization of interval decomposability may be possible.

4.2 Science of Science

I plan to expand my work in the science of science by using tools from TDA and network science to study the structure and dynamics of scientific knowledge. My work so far has focused on topological representations of scientific concepts (see Section 1.2) and scientific papers (see Section 1.3). I seek to incorporate additional entities — including authors, institutions, and their geographic locations — and to investigate the relationship between these entities and the structure of the underlying conceptual spaces. For example, one can model an author’s career as a path through a network of concepts, use PH to characterize how authors distribute themselves in relation to concepts, and study how disciplines evolve with time by tracking how papers are situated in concept space across decades. Complementing this direction, I am supervising two master’s students, Xiang Ding and Joy Yang, on a project to examine how authors move between disciplines and how these transitions relate to research impact.

4.3 Mean-Field Approaches to Collective Behavior

Agent-based models, such as those in Section 2, require Monte Carlo simulation to obtain robust statistics of model dynamics and long-term behavior. This approach is computationally expensive, and these models are difficult to study analytically. A complementary approach is to model agent-based systems from a mean-field perspective [4, 5]. For example, in opinion dynamics, this involves describing the opinion distribution with a density that evolves according to an integro-differential equation. I plan to develop mean-field counterparts to the agent-based models I have studied both to reduce computational cost and to gain analytical insights (e.g., identifying phase transitions in opinion fragmentation). I also plan to use mean-field approaches to explore systems with coupled dynamics, such as the opinion–disease system in Section 2.3.

4.4 Bridging Large-Language-Model Agent Interactions and Opinion Dynamics

Validating models of opinion dynamics with human data is difficult. Unlike physical systems, one cannot run controlled experiments at scale, nor can one access the internal states of human subjects to determine what they truly believe. Large language models (LLMs) offer a possible alternative. LLM agents can be prompted with specific instructions, interact with each other in controlled settings, and provide interpretable internal representations of their “beliefs”. They offer a useful proxy for studying opinion formation in a way that is both more tractable and more scalable than human-subject experiments.

I plan to use LLM agents to generate trajectories of opinion change and then fit opinion models to these trajectories to assess how well they capture the observed dynamics. The goodness of fit will yield insight into the specific mechanisms driving opinion change in the LLM agents. Conversely, a fitted model can serve as a proxy for understanding the dynamics of the LLM agents themselves. This opens the door to large-scale experiments that would be computationally or monetarily infeasible with LLM agents alone. For instance, one can study how introducing or removing agents while they are performing a task affects whether they successfully work together or fail to cooperate.

5 Mentoring Students

Mentoring students is central to my research and is one of the aspects that I find most rewarding. Many of my research areas offer entry points accessible to undergraduates while remaining meaningful for graduate students, allowing me to involve students at multiple levels. As a postdoc, I am supervising one undergraduate and three master’s students. As a Ph.D. student, I supervised six undergraduate research students and mentored nine students through the Directed Reading Program (DRP), as well as four students through the UCLA Applied Mathematics REU (with Mason A. Porter). Leila Thompsky and Yuexuan Wu, the first two authors on [20], were my DRP students before working on a project (co-supervised by me and Mason A. Porter) involving BCMs on networks.

Network Science There are rich opportunities for students to engage with collective dynamics on networks. Many of these projects combine theoretical analysis with numerical simulation, giving students experience in both and illustrating how they complement each other. Some projects also engage with empirical data. In opinion dynamics (see Section 2.1), students can formulate models and explore their behavior through both mathematical analysis and numerical simulations. In the science of science (see Section 4.2), students can analyze scientific advancement as dynamics on a networked system. This perspective can motivate models that seek to capture the underlying mechanisms.

Topological Data Analysis TDA offers students a powerful set of tools to uncover structure in complex data. Projects range from accessible introductions to PH on real-world data to more advanced theoretical work. In the science of science (see Section 1.2), students can use PH and Mapper to study the structure of and identify the gaps in scientific work. For students interested in geospatial applications, resource-accessibility projects (see Section 1.1) offer a concrete and intuitive entry point to TDA. Theoretically minded students can explore the foundations of TDA, such as persistence modules and their decompositions (see Section 3).

References

- [1] Gunnar Carlsson. Topology and data. *Bulletin of the American Mathematical Society*, 46(2):255–308, 2009.
- [2] Gunnar Carlsson and Vin de Silva. Zigzag persistence. *Foundations of Computational Mathematics*, 10(4):367–405, 2010.
- [3] Gunnar Carlsson and Afra Zomorodian. The theory of multidimensional persistence. *Discrete & Computational Geometry*, 42(1):71–93, 2009.
- [4] Weiqi Chu, Qin Li, and Mason A. Porter. Inference of interaction kernels in mean-field models of opinion dynamics. *SIAM Journal on Applied Mathematics*, 84(3):1096–1115, 2024.
- [5] Weiqi Chu and Mason A. Porter. A density description of a bounded-confidence model of opinion dynamics on hypergraphs. *SIAM Journal on Applied Mathematics*, 83(6):2310–2328, 2023.

- [6] Guillaume Deffuant, David Neau, Frederic Amblard, and Gérard Weisbuch. Mixing beliefs among interacting agents. *Advances in Complex Systems*, 3(01n04):87–98, 2000.
- [7] Peter Gabriel. Unzerlegbare Darstellungen I. *Manuscripta Mathematica*, 6:71–103, 1972.
- [8] Rainer Hegselmann and Ulrich Krause. Opinion dynamics and bounded confidence: Models, analysis, and simulation. *Journal of Artificial Societies and Social Simulation*, 5(3):2, 2002.
- [9] Abigail Hickok, Benjamin Jarman, Michael Johnson, **Jiajie Luo**, and Mason A. Porter. Persistent homology for resource coverage: A case study of access to polling sites. *SIAM Review*, 66(3):481–500, 2024. [AH, BJ, MJ, and JL contributed equally].
- [10] Grace J. Li, **Jiajie Luo**, and Weiqi Chu. Bounded-confidence models of multidimensional opinions with topic-weighted discordance. *SIAM Journal on Applied Dynamical Systems*, 25(1), 2026.
- [11] Grace J. Li, **Jiajie Luo**, and Mason A. Porter. Bounded-confidence models of opinion dynamics with adaptive confidence bounds. *SIAM Journal on Applied Dynamical Systems*, 24:994–1041, 2025. [GJL and JL contributed equally].
- [12] **Jiajie Luo** and James A. Evans. Filling holes in science draws collective attention, but most higher-order holes remain unexplored, 2026. arXiv:2608.28822.
- [13] **Jiajie Luo** and Gregory Henselman-Petrusek. Interval decomposition of infinite persistence modules over a principal ideal domain and field choice in persistent homology. *Proceedings of the American Mathematical Society*, 154(11):4931–4945, 2026.
- [14] **Jiajie Luo** and Gregory Henselman-Petrusek. Interval decomposition of persistence modules over a principal ideal domain. *Foundations of Computational Mathematics*, 26(4):2571–2623, 2026.
- [15] Hossein Noorazar, Kevin R. Vixie, Arghavan Talebanpour, and Yunfeng Hu. From classical to modern opinion dynamics. *International Journal of Modern Physics C*, 31(07):2050101, 2020.
- [16] Ippei Obayashi and Michio Yoshiwaki. Field choice problem in persistent homology. *Discrete & Computational Geometry*, 70(3):645–670, 2023.
- [17] Nina Otter, Mason A. Porter, Ulrike Tillmann, Peter Grindrod, and Heather A. Harrington. A roadmap for the computation of persistent homology. *EPJ Data Science*, 6:17, 2017.
- [18] Steve Y. Oudot. *Persistence Theory: From Quiver Representations to Data Analysis*, volume 209 of *Mathematical Surveys and Monographs*. American Mathematical Society, Providence, RI, 2015.
- [19] Gurjeet Singh, Facundo Mémoli, and Gunnar Carlsson. Topological methods for the analysis of high dimensional data sets and 3D object recognition. In *Eurographics Symposium on Point-Based Graphics*, pages 91–100, 2007.
- [20] Leila Thompsky, Yuexuan Yolanda Wu, Mason A. Porter, and **Jiajie Luo**. A bounded-confidence model of opinion dynamics with adaptive interaction probabilities, 2026. arXiv:2605.20418.
- [21] Sarah Tymochko, Gillian Grindstaff, Abigail Hickok, **Jiajie Luo**, and Mason A. Porter. Using persistent homology to analyze access to heterogeneous-quality resources and heterogeneous-severity nuisances, 2026. arXiv:2609.34090. [GG, AH, and JL contributed equally].